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भाग 2 विशेष अपेक्षाएं

अनुभाग 11 टम्बलर शुष्कक

Indian Standard

**SAFETY OF HOUSEHOLD AND SIMILAR
ELECTRICAL APPLIANCES**

PART 2 PARTICULAR REQUIREMENTS

Section 11 Tumbler Dryers

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Price Group 5

Electrical Appliances Sectional Committee, ETD 32

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrotechnical Division Council.

The standard covers the general and safety requirements for tumbler dryers. This standard does not cover the performance requirements which are covered under a separate standard IS 6390 (Part 2) : 1992 'Specification for domestic electrical clothes washing machine : Part 2 Requirements for tumbler dryers'.

This standard is to be read in conjunction with IS 302-1 (1979). For the sake of convenience, the clauses of this standard correspond to those of IS 302-1 (1979). Instead of reproducing full text of each clause, clauses of IS 302-1 (1979) which are applicable (which means that relevant provisions of the clause apply) or not applicable and the sub-clauses or portions of the clause apply or not applicable and the sub-clauses or portions thereof which are not applicable are indicated as under:

- a) In case of clause, where it is applicable or not applicable, the wording used is 'This clause of IS 302-1 (1979) is applicable/not applicable'; and
- b) In case of a sub-clause or part thereof 'Not applicable'.

Wherever a sub-clause of IS 302-1 (1979) is to be replaced by a new text, it has been indicated as under:

'Replacement — followed by the new text'.

Any addition to the existing provisions of a sub-clause of IS 302-1 (1979) has been indicated as under:

'Addition — followed by the text of the additional matter'.

Clauses/tables which are additional to those of IS 302-1 (1979) are numbered starting from 101 and additional sub-clauses are numbered with the main clauses number followed by 101, 102, etc, for example 7.101.

Should, however, any deviation exist between IS 302-1 (1979) and this standard, the provisions of the latter shall apply.

While preparing this standard, assistance was derived from IEC 335-2-11 (1976) 'Safety of household and similar electrical equipment : Part 2 Particular requirements, Section 11 Tumbler dryers' issued by International Electrotechnical Commission.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

The Indian Standards which are necessary adjuncts to this standard are given in Annex-AA.

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Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 11 Tumbler Dryers

1 SCOPE

This clause of IS 302-1 (1979) is applicable except as follows.

1.1 Replacement

This standard applies to electric tumbler dryers having a rated voltage not exceeding 250 V ac or dc.

Tumbler dryers incorporated in electric washing machines are within the scope of this standard.

Tumbler dryers not intended for normal household use, but which nevertheless may be a source of danger to the public, such as tumbler dryers for communal use in blocks of flats or in laundrettes, are also within the scope of this standard.

This standard does not take into account the special hazards which exist in places where there are young children or aged or infirm persons without supervision: in such cases, additional requirements may be necessary.

This standard does not apply to:

- i) Tumbler dryers designed exclusively for industrial or commercial purposes; and
- ii) Tumbler dryers intended to be used in locations where special conditions prevail, such as the presence of a corrosive or explosive atmosphere (dust, vapour or gas).

For tumbler dryers intended to be used in vehicles or on board ships or aircraft, additional requirements may be necessary.

2 TERMINOLOGY

This clause of IS 302-1 (1979) is applicable except as follows.

2.30 Replacement

Normal Load denotes the load obtained when the tumbler dryer is operated with the heating element(s) and the motor in circuit, the drum being filled with a quantity of textile

material having a mass in the dry condition equal to the maximum mass specified by the manufacturer.

The textile material is in the form of pre-washed double-hemmed cotton sheets having dimensions of approximately 70 cm × 70 cm and a mass between 140 g/m² and 175 g/m² in the dry condition.

Before the tests are started, the textile material is saturated with a quality of water having a temperature of $27 \pm 5^\circ\text{C}$ and a mass equal to that of the textile material.

For the purpose of the standard, cotton having a water content not exceeding 10 percent is considered as being the dry condition.

Cotton conditioned for 24 h in still air having a temperature of $27 \pm 2^\circ\text{C}$, a relative humidity between 60 percent and 70 percent and a pressure between 860 mbar and 1 060 mbar, will contain approximately 7 percent of water.

As an alternative to the textile material specified, pieces of cloth having an area between 4 800 cm² and 5 000 cm², with one side of at least 55 cm, may be used for the tests.

Additional Sub-Clauses

2.2.101 Tumbler dryer denotes an appliance in which textile material is dried by tumbling in a rotating drum through which heated air is blown.

2.2.102 Condensation-type tumbler dryer denotes a tumbler dryer in which the air used for the drying process is dehumidified by cooling.

3 GENERAL REQUIREMENT

This clause of IS 302-1 (1979) is applicable.

4 GENERAL NOTES ON TESTS

This clause of IS 302-1 (1979) is applicable.

5 RATING

This clause of IS 302-1 (1979) is applicable.

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6 CLASSIFICATION

This clause of IS 302-1 (1979) is applicable.

7 MARKING

This clause of IS 302-1 (1979) is applicable except as follows.

7.1 Addition

In addition, tumbler dryers shall be marked with:

- i) The maximum mass in kilogrammes, of textile material in the dry condition, for which the tumbler dryer is designed, unless this is indicated in the instruction sheet;
- ii) The subsequence of the following warnings, unless these are included in the instruction sheet:
 - a) Do not use the dryer if anything other than a water solution has been used in the cleaning process;
 - b) Clean lint trap frequently (only for tumbler dryers having means for collecting dust or lint);
 - c) Do not use heat when drying articles containing foam rubber or rubber-like materials (only for tumbler dryers incorporating a control having a 'no heat' setting);
 - d) Do not dry articles containing foam rubber or rubber-like materials (only for tumbler dryers incorporating a control not having a 'no heat' setting); and
 - e) Do not allow lint to accumulate around the dryer (not necessary for tumbler dryers intended to be vented to the exterior of the building).

Moreover, condensation-type tumbler dryers intended to be connected to the water supply mains shall be marked with:

- i) The maximum permissible inlet water pressure, in pascals, bars or newtons per square centimetre, unless this is indicated in the instruction sheet;
- ii) The minimum permissible inlet water pressure, in pascals, bars or newtons per square centimetre, if to be observed for the correct operation of the tumbler dryer, unless this is indicated in the instruction sheet.

Addition

For tumbler dryers incorporated in washing machines, which allow to dry in the container

the same quantity of textile material as put in for washing, it is not necessary to mark the maximum mass of textile material in the dry condition for which the tumbler dryer is designed.

7.10 Addition

If the 'off' position is indicated only by a word, this word shall be 'off' in the official language(s) of the country in which the tumbler dryer is to be sold.

7.12 Modification

Instead of the first sentence of the requirement, the following applies:

Tumbler dryers shall be accompanied by an instruction sheet which gives details for installation, including the type number or catalogue number of any part necessary for installation and not delivered with the tumbler dryer.

Addition

For tumbler dryers provided with ventilating opening in the base area, the instruction sheet shall include a statement drawing attention to the fact that such openings must not be obstructed by carpeting when the tumbler dryer is installed on a carpeted floor.

Additional Sub-Clauses

7.101 The tumbler dryers may also be marked with Standard Mark.

7.102 The use of the standard mark is governed by the provisions of Bureau of Indian Standards Act, 1986 and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

8 PROTECTION AGAINST ELECTRIC SHOCK

This clause of IS 302-1 (1979) is applicable.

9 STARTING OF MOTOR-OPERATED APPLIANCES

This clause of IS 302-1 (1979) is applicable.

10 INPUT AND CURRENT

This clause of IS 302-1 (1979) is applicable.

11 HEATING

This clause of IS 302-1 (1979) is applicable except as follows.

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11.2 Replacement

Tumbler dryers for building-in are built-in, dull black-painted ply-wood walls, about 20 mm thick, being used.

Other tumbler dryers are placed in a test corner. The test corner consists of two walls at right angles and a floor, all of dull black-painted plywood having a thickness of 20 mm. The tumbler dryer is positioned in the test corner as follows:

- a) Tumbler dryers normally used on a floor or a table are placed on the floor as near to the walls as possible;
- b) Tumbler dryers normally fixed to a wall are mounted on one of the walls, as near to the other wall and to the floor as is likely to occur in normal use, unless the manufacturer has given other instructions concerning their installation.

If the tumbler dryer is provided with a lint trap, this is cleaned before the test is started and its outlet opening is covered with two layers of cloth, there being no further cleaning during the test.

The composition of the cloth to be used is:

- | | |
|----------------------|--|
| a) Material | unbleached cotton |
| b) Specific mass | 25 g/m ² to 34 g/m ² |
| c) Threads per metre | |
| i) warp | 1 020 to 1 180 |
| ii) weft | 1 020 to 1 180 |

If the lint trap is of the screen type, 50 percent of the area is blocked.

11.4 and 11.5 Not applicable.

11.6 Replacement

Tumbler dryers are operated under normal load at a supply voltage such that the input is 1.15 times maximum rated input of the heating element(s) and at 1.06 times rated voltage for the motor part. If, however, the voltage required for obtaining the input specified deviates by less than +1 percent from 1.06 times rated voltage of the motor, or the heating and motor circuits cannot be operated separately, the tumbler dryer as a whole is operated at 1.06 times rated voltage.

Addition

If, for tumbler dryers incorporated in washing machines the drying function can automatically follow the washing function, the container is not separately loaded. The tumbler dryer is operated with the maximum quantity of textile material specified by the manufacturer for the combined washing-drying cycle.

11.7 Replacement

Tumbler dryers provided with a timer, a humidity sensing control or other time-limiting device are subjected to three cycles. Each cycle comprises an operating period having a duration equal to the maximum time that can be provided by the device, and a rest period of 4 min.

During the periods, the tumbler dryer is reload.

Other tumbler dryers are operated continuously until steady conditions are established.

For tumbler dryers incorporated in washing machines where the drying function automatically follows the washing function the test may be ended if, during the second cycle, the temperature rise of any part of the tumbler dryer does not exceed the value determined during the first cycle by more than 5°C.

**12 OPERATION UNDER OVERLOAD
CONDITIONS OF APPLIANCES WITH
HEATING ELEMENTS**

This clause of IS 302-1 (1979) is not applicable.

**13 ELECTRICAL INSULATION AND
LEAKAGE CURRENT AT OPERATING
TEMPERATURE**

This clause of IS 302-1 (1979) is applicable.

**14 RADIO AND TELEVISION
INTERFERENCE SUPPRESSION**

This clause of IS 302-1 (1979) is applicable.

15 MOISTURE RESISTANCE

This clause of IS 302-1 (1979) is applicable except as follows.

15.3 Replacement

Tumbler dryers shall be so constructed that spillage of liquid in normal use does not affect their electrical insulation, even in the event that an inlet valve, if any, fails to close.

Compliance is checked by the following tests:

Tumbler dryers with type X attachment are fitted with the lightest permissible type of flexible cable or cord of the smallest cross-sectional area specified in 26.2 and other tumbler dryers are tested as delivered.

Ordinary tumbler dryers are subjected to the following treatment:

A quality of 0.5 l of water, containing approximately 1 percent NaCl, is poured steadily and uniformly over the top of the tumbler dryer, with the controls in the 'on' position, so that the knobs of the controls, other than knobs

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situated above the top level of the tumbler dryer, are wetted. The controls are then operated through their full range, this operation being repeated after a period of 5 min.

Condensation-type tumbler dryers are subjected to the following additional treatment:

The drum is filled with textile material as specified for normal load, which is then saturated with an amount of water corresponding approximately with 1.5 times the maximum mass of textile material in the dry condition specified by the manufacturer.

Condensation-type tumbler dryers intended to be connected to the water supply mains are, in addition, operated with the outlet of the condensation circuit blocked. The inlet valve is held open and the filling continued for 1 min after first evidence of over-flow, or until a further protective system operates to stop the inflow. The inlet valve is still held open for 5 min after the protective system starts to operate. Immediately after this treatment, the front loading door, if any, is opened, but only if opening can be achieved without forcing inter locks or similar devices.

Immediately after these treatments, the tumbler dryer shall withstand an electric strength test as specified in 16.4 and inspection shall show that any water which may have entered the tumbler dryer does not impair compliance with this standard: in particular, there shall be no trace of water on insulation which could result in a reduction of creepage distances and clearances below the values specified in 29.1.

The tumbler dryer is allowed to stand in normal test-room atmosphere for 24 h before being subjected in the test of 15.4.

15.4 Addition

For splash-proof tumbler dryers, heating elements are subjected to the tests of 16 after the tumbler dryer has been kept for 2 days (48 h) in the humidity cabinet. The tumbler dryer is then put back in the cabinet for the remaining 5 days (120 h), after which the other parts of the tumbler dryer are tested.

16 INSULATION RESISTANCE AND ELECTRIC STRENGTH

This clause of IS 302-1 (1979) is applicable except as follows.

16.1 Replacement

The insulation and electric strength of tumbler dryers shall be adequate.

Compliance is checked by the tests of 16.2 to 16.4, which are made on the cold tumbler dryer, not connected to the supply, immediately after the test of 15.4, in the humidity cabinet or in the room in which the tumbler dryer was brought to the prescribed temperature, after reassembly of those parts which may have been removed.

16.2 Replacement

Instead of the permissible leakage current for stationary Class I appliances, the following applies:

For stationary Class I tumbler dryers, the leakage current shall not exceed 1 mA or 1 mA per kW rated input of the heating elements, whichever is the greater, with a maximum of 5 mA for the appliance as a whole.

17 OVERLOAD PROTECTION

This clause of IS 302-1 (1979) is applicable.

18 ENDURANCE (Under consideration)

19 ABNORMAL OPERATION

This clause of IS 302-1 (1979) is applicable except as follows.

19.1 Replacement

Instead of the sub-clauses applicable to the various types of appliance 19.4 to 19.8 and, if applicable, 19.10, apply.

Addition

If operation without water is considered to be a more severe condition for condensation-type tumbler dryers which are connected to the water supply. The tests are made with water to be closed. However, this tap is not closed after the tumbler dryer has been switched on.

19.2 and 19.3 — Not applicable.

19.4 Replacement

The tumbler dryer is operated under the conditions specified in 11, but with the textile material not wetted, any thermal control which operates during the test of 11 being short-circuited. The test is continued until steady conditions are established or, for tumbler dryers provided with a timer, for a period equal to the maximum period allowed by the timer.

If the tumbler dryer is provided with more than one control, these are short-circuited in turn.

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19.5 Replacement

The test of 19.4 is repeated, but with the thermal control not short-circuited and the drum-belt removed. The duration of the test is 90 min or, for tumbler dryers provided with a timer, equal to the maximum period allowed by the timer.

The test is then repeated, but with the drum-belt in position and with the air circulation stopped, if the latter condition is likely to occur due to a failure in normal use.

If both of these conditions are likely to occur simultaneously in normal use, the two tests are combined.

Care is taken to ensure that, during the latter test, the textile material in the drum is tumbling properly; if necessary, the load is suitably reduced.

19.6 Replacement

Instead of the text preceding the table, the following applies:

The tumbler dryer, starting from cold, is operated at rated voltage or at the upper limit of the rated voltage range, with the moving parts locked, for a period as long as is necessary to establishing steady conditions or, if a timer is provided, equal to the maximum period allowed by the timer.

If the motor has a capacitor in the circuit of an auxiliary winding, the test for that motor is repeated with the capacitor short-circuited or open-circuited, which is the more unfavourable unless the capacitor complies with IS 2993 : 1975.

If the tumbler dryer has more than one motor or if the motor has more than one capacitor in the circuit of an auxiliary winding, the test is made for each motor and for each capacitor separately.

Alternative tests for protected motor units are given in Annex E.

This test is made with the rotor locked because certain motors provided with capacitors might or might not start so that variable results could be obtained.

Further requirements for capacitors justifying exclusion from short-circuiting or open-circuiting are under consideration.

At the end of the test period specified, or at the instant of operation of fuses, thermal cut-outs, motor protection devices and the like, the temperature of the windings shall not exceed the values shown in the following table.

19.8 Not applicable.

19.9 Not applicable.

19.11 Addition

Moreover, the textile material in the tumbler dryer shall not ignite and shall show no charring or glowing embers before or after the door is opened.

Addition

A light-brown coloration of the textile material and a slight, emission of smoke, for example, from glowing thread ends, are ignored.

20 STABILITY AND MECHANICAL HAZARDS

This clause of IS 302-1 (1979) is applicable except as follows.

20.1 Replacement

The test with the angle of inclination or tilt increased to 15° is not made.

Additional Sub-clauses

20.101 Tumbler dryers shall be provided either with a means to prevent opening of the door while the tumbler dryer is operating as in normal use, or with an interlock which disconnects the motor before the door opening exceeds 75 mm. In the latter case, it shall not be possible to start the motor while the door opening exceeds 75 mm and, in addition, for tumbler dryers with an opening having a dimension exceeding 20 cm and a drum having a volume exceeding 100 dm³, until a separate means controlling the movement of the drum is manually operated.

Any interlock shall be so designed that unexpected operation of the tumbler dryer is unlikely to occur while the door is open.

Compliance is checked by inspection, by measurement and by manual test, the tumbler dryer being operated at rated voltage or at the upper limit of the rated voltage, and under normal load.

If a means to prevent opening of the door while the tumbler dryer is operating as in normal use incorporates a coil or similar component to hold the door locked in the closed position, this component is energized and de-energized 6 000 times, at a rate of six times per minute or at the rate imposed by the design and construction of the tumbler dryer, whichever is the lower.

During the test, no damage impairing the further use of the locking means or its components shall occur.

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If opening and closing of the door is necessary for the mechanical operation of the interlock, the door is opened and closed during the test.

Any interlock which can be released by means of the standard test finger shown in Fig. 1 of IS 302-1 (1979) is considered likely to cause unexpected operation of the tumbler dryer.

20.102 For tumbler dryers with a door opening having a dimension exceeding 20 cm and a drum volume exceeding 100 dm³, it shall be possible to open the door from the inside with a force not exceeding 70 N when applied perpendicular to the plane of the door at a point farthest from the hinges.

Compliance is checked by inspection, by measurement and by manual test.

The force may be applied at the outside of the door, at the equivalent distance from the hinges.

20.103 Tumbler dryers with horizontally hinged doors shall have adequate stability when the open door is subjected to a load.

Compliance is checked by the following test:

The tumbler dryer is placed on a horizontal surface without textile material in the container. A weight having a mass of 23 kg is then applied to the centre of open door. During the test, the tumbler dryer shall not tilt.

After the test, the door and the hinges shall not be damaged or deformed to such an extent that this will impair compliance with this standard.

For tumbler dryers which the manufacturer indicates may be placed upon another appliance, the test is not made with tumbler dryer in the stacked position.

21 MECHANICAL STRENGTH

This clause of IS 302-1 (1979) is applicable.

22 CONSTRUCTION

This clause of IS 302-1 (1979) is applicable except as follows.

22.1 Replacement

Tumbler dryers shall be of Class I, Class II or Class III.

Compliance is checked by inspection and by the relevant tests.

22.2 Replacement

Tumbler dryers shall be splash proof. Compliance is checked by inspection due by the relevant test.

22.3 Addition

If the instructions for installation indicate that the maximum permissible inclination of the supporting surface is 2°, the maximum angle of inclination is reduced to 2°.

Additional Sub-clauses

22.101 Heating elements shall be so situated or guarded that textile material cannot come into contact with them.

Compliance is checked by inspection.

22.102 For tumbler dryers incorporated in washing machines having separate containers for washing and drying, the maximum mass of textile material in the dry condition for the drying operation shall be equal to that for the washing operation.

Compliance is checked by inspection during the test of 11.

22.103 Condensation-type tumbler dryers intended to be connected to the water supply mains shall withstand the water pressure to which they may be subjected in normal use.

Compliance is checked by subjecting those parts of the tumbler dryer which are under pressure from the water supply mains, for 5 min to a static pressure equal to twice the maximum permissible inlet water pressure indicated for the tumbler dryer, or 1.2 MPa = 12 bar = 120 N/cm², whichever is the greater.

During the test, there shall be no leakage from any part, including the inlet water hose.

22.104 If the manufacturer states that the tumbler dryer can be placed on top of a washing machine, the tumbler dryer shall be supplied with a reliable means which will prevent it from tilting or falling.

Compliance is checked by inspection and by the following test:

The washing machine and tumbler dryer are assembled together in accordance with the manufacturer's instructions.

The combination is placed in all positions that deviate from the normal position of use by an angle not exceeding 5°.

Each machine is operated under normal load and supplied at rated voltage and shall operate normally without tilting.

The tumbler dryer shall not fall off the washing machine.

23 INTERNAL WIRING

This clause of IS 302-1 (1979) is applicable.

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24 COMPONENTS

This clause of IS 302-1 (1979) is applicable except as follows.

24.1 Addition

Switches incorporated in tumbler dryers are not required to be switches for frequent operation.

24.3 Addition

Switches of micro-gap construction used in connection with door interlocks are not regarded as intended to disconnect stationary appliances from the supply.

Additional Sub-clause

24.101 Thermal cut-out shall be of non-self resetting type.

25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CABLES AND CORDS

This clause of IS 302-1 (1979) is applicable except as follows.

25.1 Replacement

Tumbler dryers shall not be provided with an appliance inlet.

25.2 Replacement

Tumbler dryers shall not be provided with an appliance inlet.

25.3 and 25.10 Not applicable.

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of IS 302-1 (1979) is applicable.

27 PROVISION FOR EARTHING

This clause of IS 302-1 (1979) is applicable except as follows.

27.1 Addition

If magnetic valves have metal parts which may come into contact with accessible liquids, these parts shall either be connected to the earthing terminal or termination within the tumbler dryer, or be separated from live parts by double insulation or reinforced insulation.

28 SCREWS AND CONNECTIONS

This clause of IS 302-1 (1979) is applicable.

29 CREEPAGE DISTANCES, CLEARANCES AND DISTANCES THROUGH INSULATION

This clause of IS 302-1 (1979) is applicable.

30 RESISTANCE TO HEAT, FIRE AND TACKING

This clause of IS 302-1 (1979) is applicable.

30.4 Tumbler dryers are regarded as appliances liable to be operated while unattended.

30.5 Addition

Switches devices with moving contacts, other than those manually operated and those intended to operate only; during abnormal operation, are considered to be subjected to extra-service duty conditions.

Switches devices with moving contacts intended to operate only during abnormal; operation and other parts of insulating materials are also considered to be subjected to extra-servere duty conditions. Unless they are too enclosed or located that pollution by condensation is unlikely to occur; in such cases the requirements for service duty conditions apply.

31 RESISTANCE TO RUSTING

This clause of IS 302-1 (1979) is applicable.

32 RADIATION, TOXICITY AND SIMILAR HAZARDS

This clause of IS 302-1 (1979) is applicable except as follows.

32.1 Replacement

Test specifications are not yet available.

101 TESTS

101.1 Type Tests

The tests specified in Table 101 shall constitute the type tests and shall be carried out on two samples of the same type and rating selected preferably at random from a regular production lot. Before commencement of the tests, the samples shall be visually examined and inspected for obvious visual defects in respect of components, parts and their assembly, construction, mechanical hazards, markings, provision of suitable terminals for supply connections, earthing and the effectiveness of screws and connections. The external surface finish shall be even and free from finishing defects.

101.1.1 Criteria of Acceptance

Both samples shall successfully pass all the type tests for proving conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority at its discretion may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test(s) in which failure(s) had occurred. No failure should be permitted in the repeat test(s).

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Table 101 Schedule of Type Tests
(Clause 101.1)

Sl No.	Tests	Clause Reference
i)	Protection against electric shock	8
ii)	Starting of motor-operated appliances	9
iii)	Input	10
iv)	Temperature rise	11
v)	Operation under overload conditions	12
vi)	Electrical insulation and leakage current at operating temperature	13
vii)	Radio and television interference suppression	14
viii)	Moisture resistance	15
ix)	Insulation resistance and electric strength (after humidity treatment)	16
x)	Overload protection	17
xi)	Endurance (under consideration)	18
xii)	Abnormal operation	19
xiii)	Stability and mechanical hazards	20
xiv)	Mechanical strength	21
xv)	Construction	22
xvi)	Internal wiring	23
xvii)	Components	24
xviii)	Supply correction and external flexible cables and cords	25
xix)	Terminals for external conductors	26
xx)	Provision for earthing	27
xxi)	Screws and connections	28
xxii)	Creepage distance and clearances	29
xxiii)	Resistance to heat, fire and tacking	30
xxiv)	Resistance to rusting	31
xxv)	Radiation hazards	32

101.2 Acceptance Tests

The following shall constitute the acceptance tests.

Test

Clause Reference

- | | |
|---|----|
| a) Protection against electric shock | 8 |
| b) Input | 10 |
| c) Temperature rise | 11 |
| d) Electrical insulation and leakage current at operating temperature | 13 |
| e) Moisture resistance | 15 |
| f) Insulation resistance and electric strength (after humidity treatment) | 16 |
| g) Provision for earthing | 27 |

NOTE — For the purpose of acceptance tests, the humidity treatment shall be done for 24 hours while conducting the test for moisture resistance (15).

101.2.1 A recommended sampling procedure for acceptance tests is given in Appendix B of IS 302-1 (1979).

101.3 Routine Tests

The following shall constitute the routine tests:

Test

Clause Reference

- | | |
|--------------------------------------|-----------------------------|
| a) Protection against electric shock | 8 |
| b) High voltage | 13.3.2 of IS 302-1 (1979) |
| c) Provision for earthing | 27 |

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ANNEX A

TABLES OF TYPE TESTS

This Appendix A of IS 302-1 (1979) is not applicable.

ANNEX B

SAMPLING PROCEDURE FOR ACCEPTANCE TESTS

This Appendix B of IS 302-1 (1979) is applicable.

ANNEX C

ELECTRONIC CIRCUITS

This Appendix C of IS 302-1 (1979) is applicable if electronic circuits are used.

ANNEX D

MEASUREMENT OF TEMPERATURE WITH THERMOMETER

This Appendix D of IS 302-1 (1979) is applicable.

ANNEX E

ALTERNATIVE TESTS FOR PROTECTED MOTOR UNITS

This Appendix E of IS 302-1 (1979) is not applicable.

ANNEX F

IMPACT TEST APPARATUS

This Appendix F of IS 302-1 (1979) is applicable.

ANNEX G

THERMAL CONTROLS AND OVERLOAD RELEASES

This Appendix G of IS 302-1 (1979) is applicable if thermal controls and overload releases are used.

ANNEX H

MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES

This Appendix H of IS 302-1 (1979) is applicable.

ANNEX J

TEST FOR FIRE-RESISTING PROPERTIES

This Appendix J of IS 302-1 (1979) is applicable.

ANNEX K

**BNF JET TEST FOR DETERMINATION OF THICKNESS OF COPPER AND
NICKEL PLATING**

This Appendix K of IS 302-1 (1979) is not applicable.

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ANNEX L

**APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM ON NICKEL,
STEEL AND COPPER**

This Appendix L of IS 302-1 (1979) is not applicable.

ANNEX AA

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
302-1 (1979)	Safety of household and similar electrical appliances : Part 1 General requirements (<i>fifth revision</i>)	IS 2992 : 1975	Motor capacitors (<i>first revision</i>)

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Amendments Issued Since Publication

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